

David Kreuter

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Senior Infrastructure & Network Automation Engineer

Infrastructure automation architect with 10+ years of experience building production-grade orchestration platforms, network provisioning systems, and API-driven lifecycle management at enterprise scale. Engineered a global provisioning framework that manages 20,000+ virtual machines with automated network isolation, IP allocation, firewall policy enforcement, and multi-datacenter resilience. Deep expertise in Python middleware development, REST API integration, infrastructure-as-code (Terraform/Ansible), and security automation--including automated DMZ provisioning and Check Point firewall rule generation. Proven ability to design cloud-agnostic abstraction layers, GitOps-driven CI/CD pipelines, and high-availability architectures across geo-redundant datacenters in a regulated financial environment.

WORK EXPERIENCE

M&T Bank

03/2019 - Present

Automation Engineer (formerly Systems Engineer)

- **Network Provisioning & Lifecycle Management:** Developed an intelligent IP provisioning engine through InfoBlox API integration with automated subnet allocation and network resource lifecycle management; eliminated 1,000+ engineer hours annually and resolved a multi-year infrastructure bottleneck.
- **Network Security Automation:** Automated Check Point firewall rule generation and DMZ provisioning with programmatic security policy enforcement; integrated network isolation and a hardened security posture into the server build cycle to ensure day-zero tenant separation.
- **Platform Engineering & Middleware:** Designed and developed a custom Python orchestration application as middleware between ServiceNow and the Ansible Automation Platform; implemented advanced retry logic, automatic remediation of deployment failures, and a pub/sub queue model for high-volume, API-driven network and compute provisioning at scale.
- **Architectural Scale:** Engineered a global server provisioning framework that deployed and managed 20,000+ virtual machines; integrated Rubrik and VMware APIs to automate backups and compute placement with programmable resource lifecycle management, ensuring 100% RTO/RPO compliance at build time.
- **Infrastructure Resilience:** Engineered and maintain a highly resilient architecture spanning three identical, geo-redundant datacenters with automated failover, ensuring maximum uptime and disaster recovery readiness across distributed workloads.
- **Hybrid Cloud & GitOps:** Architected GitOps-driven CI/CD pipelines using Terraform to automate infrastructure deployments to Azure; implemented a pull-request-based workflow that standardized infrastructure changes through automated plan/apply cycles with infrastructure-as-code governance.
- **Cloud Abstraction & Redesign:** Refactored legacy on-premises automation workflows to be fully cloud-agnostic, designing an abstraction layer that enabled seamless deployment across hybrid infrastructure while maintaining a consistent programmatic interface for end users.
- **End-to-End Hardening & Provisioning:** Developed a zero-touch build workflow that delivered fully patched and hardened Linux (Ansible) and Windows (GPO) servers; automated agent installation, drive formatting, and security-compliant deployments at scale in a regulated financial environment.
- **Database as a Service (DBaaS):** Collaborated with Database Engineering to architect a high-performance, on-premises SQL cluster, enabling a unified DBaaS offering that consolidated fragmented legacy databases and streamlined application delivery.
- **Technical Leadership:** Serve as a primary mentor for the Technology Development Program (TDP), setting enterprise-wide standards for naming conventions, Git branching strategies, and unit testing for infrastructure code.
- **Cost Optimization:** Orchestrated the migration of 850 branch file servers within a strict three-month window using custom PowerShell automation, avoiding \$400,000 in extended licensing fees.
- **Risk Mitigation:** Authored Active Directory lifecycle scripts with built-in read-only dry-run modes; implemented least-privilege permissions and robust rollback protocols for stale object cleanup.

OTG Management

07/2013 - 02/2019

Lead Network Engineer

- **Infrastructure Resilience:** Architected a highly redundant DNS solution that significantly reduced network outages and optimized global traffic performance across distributed infrastructure.
- **Cloud Connectivity:** Established secure, encrypted VPN tunnels to bridge on-premises networks with cloud resources, ensuring seamless hybrid-cloud operations and network isolation.
- **Performance Engineering:** Diagnosed and resolved a mission-critical server hang by identifying a NUMA architecture mismatch; realigned vCPU/memory topology to eliminate cross-node latency and restore system stability.
- **Security Architecture:** Engineered a secure reverse proxy solution to protect sensitive internal infrastructure, enabling customer online ordering while mitigating external threat vectors.

Network Engineer

- **Network Administration:** Managed complex firewall rules, switch ACLs, and network policies for a massive distributed network spanning 12 airports and 3 satellite offices, ensuring tenant isolation and security compliance.
- **Proactive Monitoring:** Developed service-aware alerting and telemetry frameworks for business-critical systems, providing

network observability and reducing mean time to resolution (MTTR) through proactive incident detection.

- **Cybersecurity Governance:** Directed enterprise-wide vulnerability scanning and remediation efforts for internal and external systems to maintain a hardened security posture.
- **Enterprise Mobility:** Successfully onboarded and managed 15,000 iPads via Meraki MDM, implementing automated security protocols including geofencing, remote-wipe triggers, and custom application deployment.

Systems Administrator

- **Operational Visibility:** Modernized infrastructure monitoring by designing custom Grafana dashboards to visualize real-time performance metrics and network telemetry.
- **Disaster Recovery:** Spearheaded the deployment of Datto backup and disaster recovery appliances to ensure 100% business continuity for critical operations.

SKILLS

Languages: Python, Go (familiar), Bash, SQL, PowerShell, YAML, Javascript, PHP

Networking: BGP, DNS, VXLAN, VPN, F5 Load Balancing, High Availability, VRRP, Intrusion Detection, Vulnerability Scanning, Meraki MDM

Automation/Orchestration: Terraform, REST API, GitOps, CI/CD Pipelines, Ansible Automation Platform, Middleware Development, Gitlab, Grafana, GraphQL, ServiceNow, Unit Testing

Infrastructure: Linux Networking, VMware vSphere, Azure Cloud Services, Check Point Firewalls, Infrastructure as Code (IaC), OS Hardening (CIS/STIG), DBaaS, NUMA Tuning, Nutanix AHV, Rubrik, Active Directory

EDUCATION

Bachelor Of Science in Computer Information Systems

12/2012

SUNY Fredonia

CERTIFICATIONS

Certified Meraki Network Operator

Meraki

VOLUNTEERING & LEADERSHIP

Habitat For Humanity

07/2025 - Present

Volunteer

Tift Nature Preserve

01/2025 - Present

Volunteer

716 Paws

01/2022 - Present

Thermal drone pilot for search and rescue missions.